AWS Elastic File System (EFS) Project Guide

Overview

Amazon EFS provides a scalable, fully managed elastic NFS file system for use with AWS Cloud services and on-premises resources. This guide documents the complete EFS setup with IAM permissions and network configuration.

Architecture Diagram

```
┌─────────────────────────────────────────────────────────────────────┐
│                      AWS CLOUD (us-west-1 N. California)              │
│                                                                       │
│  ┌──────────────────────────────────────────────────────────────┐    │
│  │                   VPC: Default (vpc-08cc0f47e139b54df)         │    │
│  │                        CIDR: 172.31.0.0/16                      │    │
│  │                                                                 │    │
│  │    ┌──────────────────────────────────────────────────────┐   │    │
│  │    │              EFS: mysystem (fs-029ad906463130926)     │   │    │
│  │    │                   Encrypted: ✅ Yes                     │   │    │
│  │    │                   Size: 6.00 KiB                       │   │    │
│  │    └──────────────────────────────────────────────────────┘   │    │
│  │                   │                              │              │    │
│  │          ┌────────┴────────┐           ┌────────┴────────┐     │    │
│  │          │ Mount Target 1  │           │ Mount Target 2  │     │    │
│  │          │ AZ: us-west-1a  │           │ AZ: us-west-1c  │     │    │
│  │          │ IP: 172.31.25.41│           │ IP: 172.31.3.34 │     │    │
│  │          │ subnet-04ae2ec  │           │ subnet-09ced74  │     │    │
│  │          └────────┬────────┘           └────────┬────────┘     │    │
│  │                   │                             │               │    │
│  │          ┌────────┴─────────────────────────────┴──────┐       │    │
│  │          │        Security Group: EFS-SG                │       │    │
│  │          │        (sg-00ca265c2656fd3e2)                │       │    │
│  │          │        Allows NFS traffic (Port 2049)        │       │    │
│  │          └──────────────────────────────────────────────┘       │    │
│  │                                                                 │    │
│  │      ┌─────────────┐      ┌─────────────┐      ┌─────────────┐  │    │
│  │      │ EC2 Instance│      │ EC2 Instance│      │ EC2 Instance│  │    │
│  │      │ (AZ: 1a)    │      │ (AZ: 1c)    │      │ (Any AZ)    │  │    │
│  │      │ Mount EFS   │      │ Mount EFS   │      │ Mount EFS   │  │    │
│  │      └─────────────┘      └─────────────┘      └─────────────┘  │    │
│  │                                                                  │    │
│  └──────────────────────────────────────────────────────────────────┘    │
│                                                                           │
│  ┌──────────────────────────────────────────────────────────────┐        │
│  │                     IAM Configuration                         │        │
│  │  User: IAM_Punith                                              │        │
│  │  Policies: EC2FullAccess, EFSFullAccess, VPCFullAccess, IAMFullAccess │ │
│  └──────────────────────────────────────────────────────────────┘        │
│                                                                           │
│ Other VPCs Available:                                                     │
```

```
│  ├── MY cloud-vpc (vpc-001c72ca44f09fb81) - CIDR: 10.x.x.x                      │
│  └── on premises-vpc (vpc-0cf4fb1834013a019) - CIDR: 172.x.x.x                  │
│                                                                                   │
└───────────────────────────────────────────────────────────────────────────────────┘
```

Project Components

Component	Value	Description
EFS Name	mysystem	Elastic File System for shared storage
File System ID	fs-029ad906463130926	Unique identifier
Region	us-west-1 (N. California)	AWS Region
Encryption	Enabled	Data encrypted at rest
IAM User	IAM_Punith	User with EFS management permissions
VPC	vpc-08cc0f47e139b54df	Default VPC for EFS
Security Group	EFS-SG (sg-00ca265c2656fd3e2)	Controls NFS access

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Prerequisites

- AWS Account with appropriate permissions
- Understanding of VPC, Subnets, and Security Groups
- EC2 instances to mount EFS

Step 1: Create IAM User with Required Permissions

> *IAM user needs proper policies to manage EFS and related resources.*

1.1 Create IAM User

1. Go to **IAM Console → Users → Create user**
2. Enter username: `IAM_Punith`
3. Enable **AWS Management Console access**
4. Click **Next**

1.2 Attach Required Policies

Policy Name	Type	Purpose
`AmazonEC2FullAccess`	AWS Managed	Manage EC2 instances to mount EFS
`AmazonElasticFileSystemFullAccess`	AWS Managed	Full EFS management
`AmazonVPCFullAccess`	AWS Managed	Manage VPC and networking
`IAMFullAccess`	AWS Managed	Manage IAM resources
`EC2InstanceConnect`	AWS Managed	Connect to EC2 via browser

`IAMUserSSHKeys`	AWS Managed	Manage SSH keys
`AmazonConnect_FullAccess`	AWS Managed	Amazon Connect service

Your IAM User Setup:

```
User: IAM_Punith
├── AmazonConnect_FullAccess (AWS Managed) - Directly Attached
├── AmazonEC2FullAccess (AWS Managed) - Directly Attached
├── AmazonElasticFileSystemFullAccess (AWS Managed) - Directly Attached
├── AmazonVPCFullAccess (AWS Managed) - Directly Attached
├── EC2InstanceConnect (AWS Managed) - Directly Attached
├── IAMFullAccess (AWS Managed) - Directly Attached
└── IAMUserSSHKeys (AWS Managed) - Directly Attached
```

Step 2: Create Security Group for EFS

> *Security group controls which instances can access the EFS file system.*

1. Go to **EC2 Console** → **Security Groups** → **Create Security Group**
2. Configure:

Setting	Value
Name	`EFS-SG`
Description	Security group for EFS mount targets
VPC	Default VPC (vpc-08cc0f47e139b54df)

3. **Inbound Rules:**

Type	Protocol	Port	Source	Description
NFS	TCP	2049	0.0.0.0/0 or specific SG	Allow NFS traffic

4. Click **Create Security Group**

Your Security Group:

```
Security Group: EFS-SG
ID: sg-00ca265c2656fd3e2
VPC: vpc-08cc0f47e139b54df (default)
Inbound: NFS (TCP 2049) from allowed sources
```

Step 3: Create EFS File System

1. Go to **EFS Console** → **Create file system**
2. Click **Customize** for advanced options
3. Configure:

General Settings:

Setting	Value
Name	`mysystem`
Storage class	Standard
Automatic backups	Enable/Disable as needed
Lifecycle management	Configure as needed
Encryption	✅ **Enable encryption at rest**

4. Click **Next**

Network Settings:

Setting	Value
VPC	Default VPC (vpc-08cc0f47e139b54df)

5. Click **Next** → **Create**

Your EFS File System:

```
Name: mysystem
File System ID: fs-029ad906463130926
Encrypted: ✅ Yes
Total Size: 6.00 KiB
Size in Standard: 6.00 KiB
Size in IA: 0 Bytes
Size in Archive: 0 Bytes
State: Available
```

Step 4: Configure Mount Targets

> *Mount targets allow EC2 instances in specific AZs to connect to EFS.*

1. Go to **EFS Console** → Select `mysystem` → **Network** tab
2. Click **Manage**
3. Configure mount targets for each Availability Zone:

Mount Target Configuration:

Availability Zone	Subnet ID	IP Address	Security Group
us-west-1a	subnet-04ae2ec	172.31.25.41	EFS-SG (sg-00ca265c2656fd3e2)
us-west-1c	subnet-09ced74	172.31.3.34	EFS-SG (sg-00ca265c2656fd3e2)

4. Select **EFS-SG** security group for both mount targets
5. Click **Save**

Your Mount Targets:

```
Mount Target 1:
├── AZ: us-west-1a
├── Subnet: subnet-04ae2ec
├── IP: 172.31.25.41
├── IP Type: IPv4 only
└── Security Group: EFS-SG

Mount Target 2:
├── AZ: us-west-1c
├── Subnet: subnet-09ced74
├── IP: 172.31.3.34
├── IP Type: IPv4 only
└── Security Group: EFS-SG
```

Step 5: VPC Configuration

Your Available VPCs:

VPC Name	VPC ID	State	CIDR Block
MY cloud-vpc	vpc-001c72ca44f09fb81	Available	10.x.x.x/16
on premises-vpc	vpc-0cf4fb1834013a019	Available	172.x.x.x/16
Default (no name)	vpc-08cc0f47e139b54df	Available	172.31.0.0/16

> ***Note:*** *EFS is configured in the Default VPC. If you need EFS access from other VPCs, you can use VPC Peering or Transit Gateway.*

Step 6: Mount EFS on EC2 Instance

6.1 Launch EC2 Instance

1. Launch an EC2 instance in the same VPC as EFS
2. Ensure the instance security group allows outbound NFS (port 2049)

6.2 Install EFS Mount Helper

```bash
# For Amazon Linux 2
sudo yum install -y amazon-efs-utils

# For Ubuntu
sudo apt-get update
sudo apt-get install -y amazon-efs-utils
```

6.3 Create Mount Point

```
sudo mkdir /mnt/efs
```

6.4 Mount EFS Using EFS Mount Helper (Recommended)

```
# Mount with encryption in transit
sudo mount -t efs -o tls fs-029ad906463130926:/ /mnt/efs

# OR without encryption
sudo mount -t efs fs-029ad906463130926:/ /mnt/efs
```

6.5 Mount EFS Using NFS Client

```
# Install NFS client
sudo yum install -y nfs-utils  # Amazon Linux
sudo apt-get install -y nfs-common  # Ubuntu

# Mount using NFS
sudo mount -t nfs4 -o
nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport fs-
029ad906463130926.efs.us-west-1.amazonaws.com:/ /mnt/efs
```

6.6 Verify Mount

```
df -h /mnt/efs
```

6.7 Auto-Mount on Reboot

Add to `/etc/fstab`:

```
# Using EFS mount helper
fs-029ad906463130926:/ /mnt/efs efs defaults,_netdev 0 0

# Using NFS
fs-029ad906463130926.efs.us-west-1.amazonaws.com:/ /mnt/efs nfs4
nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport,_netdev 0 0
```

Step 7: Test EFS

Create Test File

```
# On Instance 1
sudo touch /mnt/efs/test-file.txt
echo "Hello from Instance 1" | sudo tee /mnt/efs/test-file.txt
```

Verify on Another Instance

```
# On Instance 2 (same EFS mounted)
cat /mnt/efs/test-file.txt
# Output: Hello from Instance 1
```

Troubleshooting

Issue	Solution
Mount timeout	Check security group allows NFS (2049) inbound
Permission denied	Verify IAM permissions and security groups
DNS resolution failed	Ensure VPC DNS hostnames enabled
Mount helper not found	Install amazon-efs-utils package

Useful Commands:

```
# Check EFS mount status
mount | grep efs

# Check NFS connections
netstat -an | grep 2049

# Debug mount issues
sudo mount -t efs -o tls,mounttargetip=172.31.25.41 fs-029ad906463130926:/ /mnt/efs

# Unmount EFS
sudo umount /mnt/efs
```

Cleanup / Delete Resources

> ***Important:*** *Delete resources in the correct order.*

Deletion Order:

Step	Resource	Action
1	Unmount EFS	Unmount from all EC2 instances
2	EC2 Instances	Terminate instances (optional)
3	EFS File System	Delete the file system
4	Security Group	Delete EFS-SG
5	IAM User	Delete user (if no longer needed)

Delete EFS:

1. **EFS Console** → Select `mysystem`
2. Click **Delete**
3. Enter file system ID to confirm: `fs-029ad906463130926`

Summary

Step	Action
1	Create IAM User (IAM_Punith) with required policies
2	Create Security Group (EFS-SG) allowing NFS traffic
3	Create EFS File System (mysystem) with encryption
4	Configure Mount Targets in multiple AZs
5	Configure VPC networking
6	Mount EFS on EC2 instances
7	Test file sharing across instances

Key Information Reference

```
┌──────────────────────────────────────────────────────────────┐
│                    QUICK REFERENCE                           │
├──────────────────────────────────────────────────────────────┤
│ EFS File System ID : fs-029ad906463130926                    │
│ EFS Name           : mysystem                                │
│ Region             : us-west-1 (N. California)               │
│ VPC                : vpc-08cc0f47e139b54df                   │
│ Security Group     : sg-00ca265c2656fd3e2 (EFS-SG)           │
│ IAM User           : IAM_Punith                              │
│                                                              │
│ Mount Target 1 (us-west-1a): 172.31.25.41                    │
│ Mount Target 2 (us-west-1c): 172.31.3.34                     │
│                                                              │
│ Mount Command:                                               │
│ sudo mount -t efs fs-029ad906463130926:/ /mnt/efs            │
│                                                              │
│ DNS Name:                                                    │
│ fs-029ad906463130926.efs.us-west-1.amazonaws.com             │
└──────────────────────────────────────────────────────────────┘
```

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